

Stage One Technical Report: Adaptive Memory Dynamics

Reproducible comparison of persistent, passive-decay, and reinforced-retention mechanisms

Core Research Document - Frozen evidence report

Adaptive Memory Dynamics Research Program Stage One revision - September 2026

Evidence status: Stage One is a reproducible sklearn-digits-8x8 experiment across five independent seeds. Reinforced retention approximately matches the persistent control on ordinary accuracy and Task B adaptation; passive decay reduces accuracy; neither decay mechanism improves Task A retention in the current sequential setup.

Executive summary

Stage One implements a reproducible three-condition memory experiment on sklearn-digits-8x8 with five independent seeds (7, 21, 42, 84, 101), 24 supervised epochs, and 16 epochs per continual-learning task. Reinforced retention approximately matches the persistent control on ordinary test accuracy and Task B adaptation. Passive decay reduces accuracy. Most importantly, neither decay mechanism improves Task A retention: the persistent baseline forgets less. This result validates the experimental pipeline while falsifying the strongest immediate version of the hypothesis that the current decay rules mitigate catastrophic forgetting.

1. Frozen configuration

Item	Stage One setting
Dataset	sklearn-digits-8x8
Seeds	7, 21, 42, 84, 101
Supervised training	24 epochs
Continual training	16 epochs per task
Mechanisms	persistent; passive decay; reinforced retention
Controls	identical model/training settings across mechanisms

2. Aggregate results

Mechanism	Final test	Post-idle	Task B	Task A forgetting
Persistent	0.9618 ± 0.0092	0.9618 ± 0.0092	0.9741 ± 0.0037	0.6947 ± 0.1014
Passive decay	0.9480 ± 0.0088	0.9458 ± 0.0051	0.9661 ± 0.0051	0.9858 ± 0.0058
Reinforced retention	0.9622 ± 0.0047	0.9613 ± 0.0058	0.9759 ± 0.0024	0.8619 ± 0.0558

Figure 1. Measured mean accuracy ± sample SD across the five frozen seeds.

Figure 2. Measured Task A forgetting ± sample SD. Lower is better.

3. Interpretation

Reinforced retention is competitive with persistent retention on ordinary held-out accuracy and Task B adaptation in this small benchmark, but this does not translate into superior preservation of Task A. Passive decay is worse on ordinary accuracy and prior-task retention in the frozen configuration. The evidence therefore supports mechanism differentiation and pipeline validity, not a general performance advantage for forgetting.

4. What Stage One establishes

- The repository can run the three mechanisms under matched settings across repeated seeds.
- The export pipeline produces raw epoch metrics, run summaries, manifests, and figures.
- The mechanisms produce measurably different retention/accuracy behavior in the reference experiment.
- A negative result is preserved rather than tuned away.

5. What Stage One does not establish

- No claim that adaptive forgetting solves catastrophic forgetting.
- No evidence yet on the capstone's planned MNIST, Fashion-MNIST, CIFAR-10, or larger continual-learning benchmarks.
- No biological validation of the memory analogy.
- No evidence that the current default decay/reinforcement hyperparameters are optimal.
- No architecture-general conclusion.

6. Threats to validity

- Small digits benchmark may not represent larger visual or continual-learning tasks.
- Five seeds are useful preliminary replication but may be insufficient for small effects.
- A single task partition/order can confound mechanism effects with sequence geometry.
- The forgetting metric must be interpreted alongside starting Task A competence and adaptation performance.
- Hyperparameter defaults can strongly influence whether decay appears useful or destructive.

7. Next experiment

Run rate/strength/threshold ablations, multiple task orders, and larger benchmark replication. Freeze discovery settings before confirmatory evaluation and retain the persistent baseline in every block.

Figures

